

CLEANALERT: SMART WASTE COLLECTION NOTIFICATION AND COMMUNITY REWARD SYSTEM

ABSTRACT

CleanAlert is an Android-based smart waste management application designed to improve communication between residents, waste collectors, and administrators. In many localities, residents may not know when a collection vehicle is approaching, while collection teams and administrators may lack a shared digital view of submissions, routes, and completed work. CleanAlert brings these workflows together through role-based screens, photo-based waste submissions, route assignment, collection history, and a community reward mechanism. The resident workflow supports account registration and login, selecting a waste category, attaching a photograph, viewing submission history, and checking points and rewards. The collector workflow provides a dashboard, availability status, assigned route information, a map view, and collection history. The administrator workflow includes user and collector management, route assignment, submission review, reward catalogue management, and reports. The project is designed around Kotlin and Jetpack Compose for the Android interface, Firebase Authentication for account access, Cloud Firestore for application data, Firebase Storage for submitted images, Firebase Cloud Messaging for notifications, and Google Maps and Android location services for map-related functionality. The rewards workflow is intended to encourage verified participation in responsible waste disposal. This report presents the project overview, related concepts, system design, requirements, methodology, implementation modules, testing plan, and future scope. The screenshots demonstrate the planned interface structure; however, live GPS updates, notification delivery, backend security, and reward accounting must be validated through end-to-end testing before operational claims can be made.

Keywords: Smart waste management, Android, GPS, Firebase, Google Maps, waste collection, community rewards, notifications.

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